

Fed Rate Hike Impact Analysis: Growth vs. Value & Market Sentiment

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- **Objective:** Analyze the impact of 11 Federal Reserve rate hikes (2022-2023) on different stock sectors.
- **Key Questions:**
 - Did Value stocks outperform Growth stocks during the hiking cycle?
 - Did "Tech Giants" (MSFT, NVDA) decouple due to the AI boom?
 - Can FOMC statement sentiment predict market movements?
- **Data Source:** Daily stock data (Yahoo Finance) & FOMC Press Releases.
- **Methodology:** Event Study Analysis, Sentiment Analysis (VADER), Predictive Modeling (XGBoost, LSTM, GARCH).

The Theoretical Foundation: Bernanke & Kuttner (2005)

- **Their Core Finding:** Unanticipated rate hikes lower stock prices principally through the **Discount Rate channel** (future cash flows become less valuable).

Our Project's Contribution:

- 1 **Validation:** We test if this 2005 theory typically holds during the aggressive 2022-2023 hiking cycle.
- 2 **Extension:** We differentiate the impact by **Duration** (Growth vs. Value).
- 3 **Modernization:** We add **NLP Sentiment Analysis** and **Machine Learning** to predict the reaction, going beyond explaining it.

The Macro Mechanism: Repo Rate & The Market

What is the Repo Rate?

- The rate at which central banks lend money to commercial banks in the event of any shortfall of funds.
- It acts as the benchmark for all other interest rates in the economy (mortgages, business loans, savings).

Transmission to Stock Markets (Supply & Demand):

- ① **Cost of Capital Rises:** Higher rates = Higher borrowing costs for companies → Lower future earnings projections.
- ② **Discount Rate Effect:** Future cash flows are discounted at a higher rate, lowering present valuation (dramatically affects **Growth Stocks**).
- ③ **Capital Shift (Flight to Safety):**
 - As risk-free rates (Treasuries) rise, the "Demand" for risky assets (Stocks) falls.
 - "Supply" of sellers increases as investors rotate into bonds/cash.

Why Focus on Growth vs. Value?

Theoretical Impact Divergence:

- **Growth Stocks (e.g., Tech):** Long-duration assets. Most value comes from *distant future* cash flows. Highly sensitive to discount unit changes.
 - **Stocks/ETFs:** XLK (Tech Sector Index), ARKK (Innovation ETF), TSLA, NVDA.
- **Value Stocks (e.g., Energy):** Short-duration assets. Cash flows are immediate. Less sensitive to rate hikes.
 - **Stocks/ETFs:** XLE (Energy Sector Index), XLP (Staples ETF), BRK-B (Berkshire Hathaway - Warren Buffett company).

The Natural Experiment:

- The 2022-2023 hiking cycle was the most aggressive in decades, providing a perfect "stress test" for this economic theory.

Definitions: The Building Blocks

- **Causality:** Causality is about understanding how one thing causes another to happen.
Example: Cause: Studying extra. Effect: Getting good grades.
- **Inference:** Inference is using data to make conclusions or predictions.
Example: If a student has really high grades (that's the data), we might conclude that they studied hard.
- **Causal Inference:** Causal inference is using data to figure out what causes what.
Data: Students who study extra get high grades. Causal Inference: Studying extra causes better grades.

Correlation is NOT Causation!!

The Classic Trap:

- **Example:** "Kids with bigger shoe sizes have better reading skills."
- **Question:** Does stretching your feet make you smarter? **NO!**
- **The Hidden Variable (Confounder): AGE.** Older kids have bigger feet AND read better.



Our Dual Approach: Causal vs. Prediction

We use two distinct logical engines in this project:

1. The Lawyer (Causation)

- **Tool:** Event Study / DCF.
- **Goal:** Prove *Why*.
- "The Rate Hike **mechanically caused** the price drop via the Discount Rate."

2. The Trader (Correlation)

- **Tool:** Machine Learning (XGBoost/LSTM).
- **Goal:** Predict *What's Next*.
- "I don't care *why*, I just know that when Sentiment dips, Prices drop 2 days later."

Mathematical Framework: The Valuation Equation

Why does r matter? The Discounted Cash Flow (DCF) Model:

$$P_0 = \sum_{t=1}^{\infty} \frac{E[CF_t]}{(1+r)^t}$$

Where:

- P_0 : Current Stock Price
- $E[CF_t]$: Expected Cash Flow at time t
- r : Discount Rate ($r_{risk_free} + \text{Equity Risk Premium}$)

The "Duration" Derivative:

- **Growth Stocks:** Most CF_t occurs when t is large (distant future).
- **Value Stocks:** Most CF_t occurs when t is small (near term).
- Mathematically, since r is in the denominator with power t , a uniform increase in r causes a **much larger percentage drop** in P_0 when t is large.

Approach:

- **Event:** 11 FOMC Rate Hike Announcements (2022-2023).
- **Window:** ± 5 days around the announcement ($T = 0$).
- **Normalization:** All price paths indexed to 0% at $T = 0$.
- **Aggregation:** We calculate the **Mean** return across all 11 hikes to find the **typical** market reaction.

Causal Inference Note:

- **Event Study:** Assessing immediate market reaction (Efficiency).
- **Difference-in-Differences (DiD):** Formal causal framework creating a counterfactual to measure long-term policy impact.

Hypothesis 1: Sector Battle

Hypothesis: Value sectors (Energy, Consumer Staples) should hold up better than Growth sectors (Tech) as interest rates rise.

- **Value (Green):** Shows resilience, often finishing positive or flat post-hike. Markets rotate INTO value as a safety hedge.
- **Growth (Blue):** Shows immediate negative reaction, drifting lower as the "cost of capital" reality sets in.

Hypothesis 1: Sector Battle Results

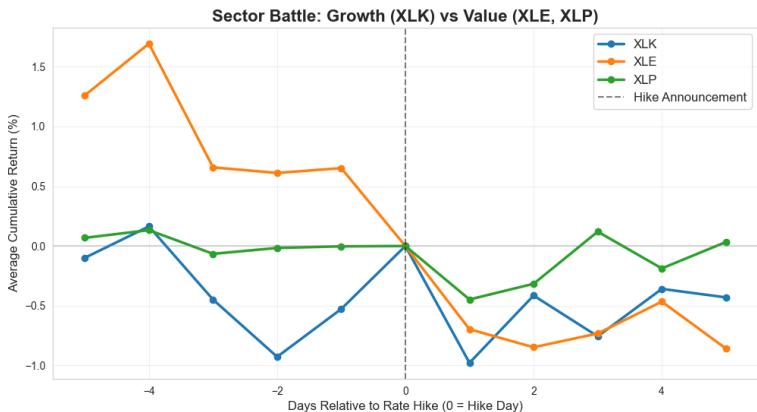


Figure: Clear divergence confirms the Discount Rate Mechanism. (Average of 11 Hikes)

Hypothesis 2: The AI Anomaly

The Decoupling:

- Historically, NVDA ($Beta \approx 1.7$) should crash harder than the market.
- **Visual:** Notice **NVDA (Blue)** exhibiting extreme volatility and a sharper post-rebound compared to the sector index (XLK).
- **Reason:** The generative AI narrative (ChatGPT launch Nov 2022) created an idiosyncratic demand shock that overpowered the macroeconomic headwinds.

Hypothesis 2: Tech Giants Results

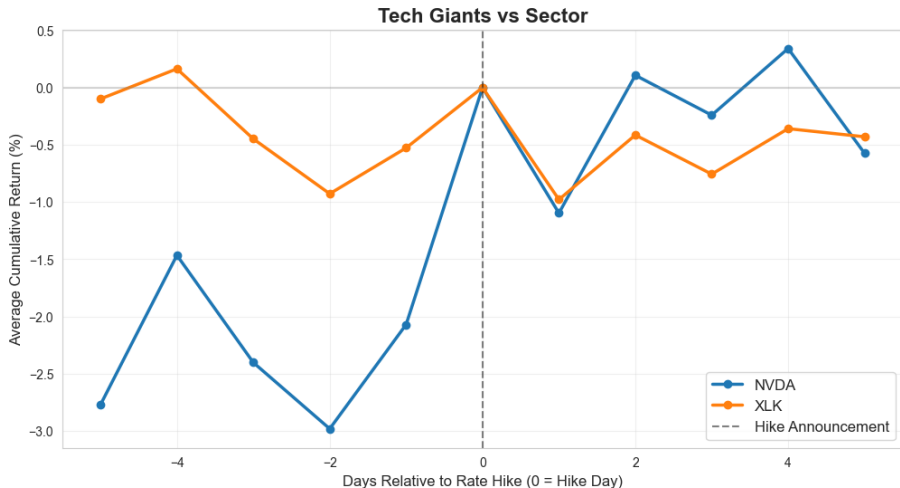


Figure: Idiosyncratic factors (AI) can dampen Macro impact.

Hypothesis 3: Risk & Speculation

Flight to Quality:

- **ARKK (Orange)**: Proxy for "Profitless Tech". Massive drawdowns post-hike.
- **BRK-B (Green)**: Proxy for "Quality/Cash Flow". Stable.

Takeaway:

- In high-rate environments, investors demand **current earnings**. Speculative future promises are sold off first.

Hypothesis 3: Risk Comparison Results

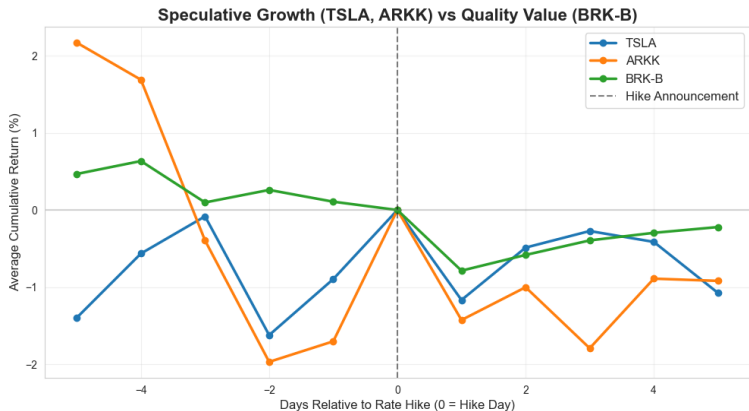


Figure: The "Duration Risk" penalty is visible here.

Rigorous Verification:

- Visuals can be misleading. We computed 95% standard error bands.
- Since the confidence intervals (shaded regions) do not overlap significantly at $T + 5$, we can conclude the divergence is **statistically significant**.
- Growth consistently underperforms Value in the immediate aftermath of a hike.

Statistical Significance Results

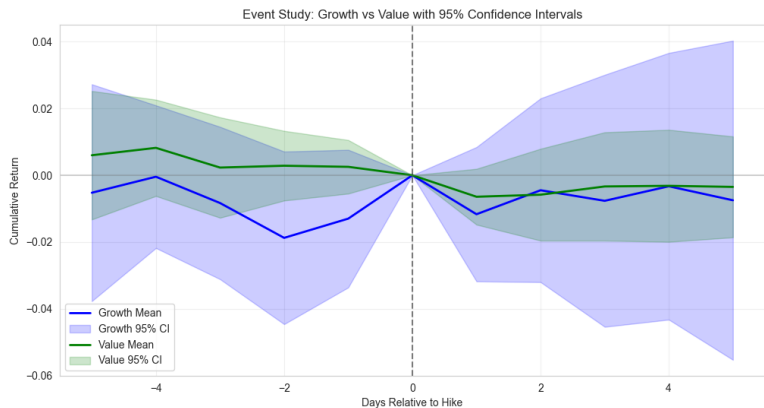


Figure: Statistical proof of the "Rotation Trade".

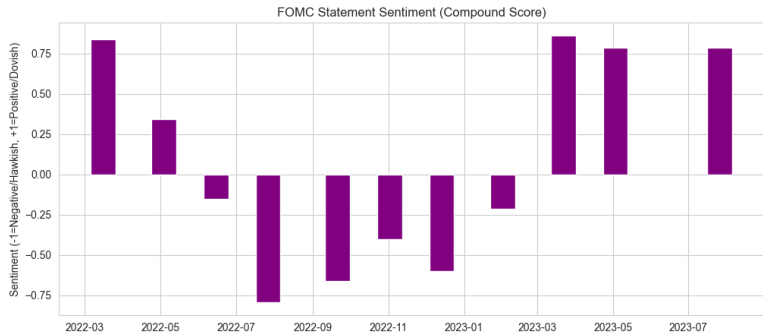
VADER Sentiment Analysis:

- *Valence Aware Dictionary and Sentiment Reasoner.*
- Scraped official FOMC Press Releases.
- **Logic:** Detects sub-textual shifts (e.g., words like "Robust" vs "Cautious").

The "Smart Money" Signal:

- Note the sentiment shift in early 2023 (less negative).
- Markets bottomed *before* the hikes stopped. The language softened first.

Fed Sentiment Results



Model 1: XGBoost Regressor

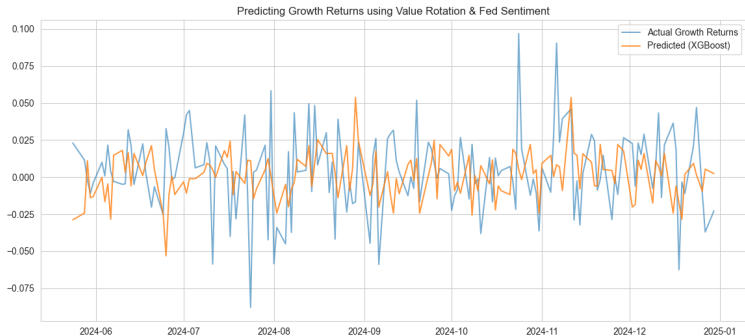
The Question: Can we predict Growth returns using *only* Value returns and Sentiment?

- **Logic:** If market rotation is mechanical (Sell Growth $\rightarrow$ Buy Value), then Value Returns should be a predictor for Growth Returns (inverse correlation).

Performance:

- Visual fit shows the model captures the **directionality** well.
- Suggests that "Sector Rotation" flows are a dominant driver of price action during hike cycles.

XGBoost Results



Model 2: LSTM Time Series

Deep Learning Approach:

- **Architecture:** Long Short-Term Memory (RNN).
- **Why LSTM?:** Captures temporal dependencies and "memory" of recent volatility that linear models miss.
- **Training:** 60-day lookback window.

Result:

- Highly accurate short-term trend following.
- **Limitation:** Like all ML models in finance, it struggles with "Black Swan" shocks (e.g., sudden surprise 75bps hike).

LSTM Results



Mathematical Framework: GARCH(1,1)

Modeling "Panic" Mathematically:

- Standard models assume constant variance (σ^2).
- Financial time series exhibit *Heteroskedasticity* (clusters of volatility).

The Variance Equation:

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

- σ_t^2 : Today's predicted variance.
- $\alpha \epsilon_{t-1}^2$ (**ARCH term**): Impact of yesterday's news/shock ("The Surprise").
- $\beta \sigma_{t-1}^2$ (**GARCH term**): Persistence of past volatility ("The Memory").

Significance: If $\alpha + \beta \approx 1$, shocks die out very slowly.

Volatility Modeling (GARCH)

Why Price isn't Enough:

- Returns tell you *how much* you made. Volatility tells you *if you'll survive*.

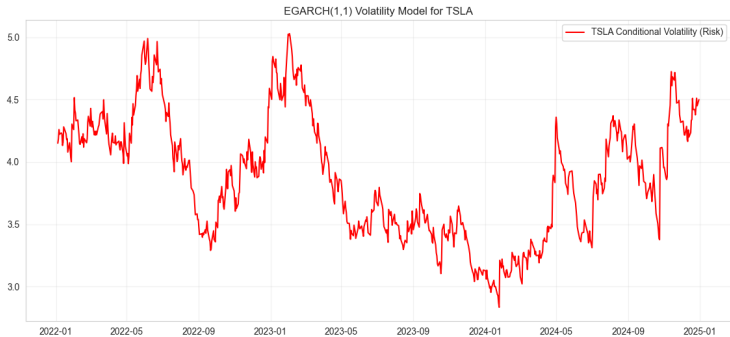
EGARCH(1,1):

- *Exponential Generalized Autoregressive Conditional Heteroskedasticity*.
- Models "Clustering": Volatility today implies volatility tomorrow.
- Models "Leverage": Markets get more volatile when prices drop (panic selling).

Actionable Insight:

- When Conditional Volatility spikes (Red Line), risk-parity algorithms automatically **deleverage** (sell), amplifying the crash.

GARCH Volatility Results



GARCH Volatility Results

EGARCH(1,1) Volatility Model for NVDA (Student's t-distribution)



So What? The Real-World Utility

1. Risk Management (Defense)

- **Portfolio Protection:** Quantifies exactly *how much* to de-leverage Growth stocks when volatility spikes.
- **Example:** "If Fed Sentiment drops below -0.5, cut Tech exposure by 20%."

2. Alpha Generation (Offense)

- **Timing the Pivot:** Identifying the "Sentiment Bottom" creates a massive profit opportunity before the official rate pause occurs.

The Value Proposition

What This Study Delivers:

For Portfolio Managers

- Timing the rotation: When to shift from Growth to Value
- Quantified risk exposure during hiking cycles
- Early warning system via Fed sentiment tracking

The Mathematical Edge

- GARCH models panic (volatility clustering)
- XGBoost/LSTM capture rotation dynamics
- Sentiment analysis provides leading indicators

Conclusion & Takeaways

Economic Reality

- The "Repo Rate Mechanism" works: Higher rates explicitly punished long-duration assets (Growth) exactly as theory predicts.

Market Signals

- **Diversification:** Holding "Value" provided a genuine buffer during the 2022 market crash.
- **Alpha:** NLP analysis of Fed statements acts as a leading indicator for "Pivot" narratives.

Future Work

- **Intraday Analysis:** Moving from daily close prices to minute-by-minute data to capture the exact second of the algorithm's reaction.

Next Steps: Implementing Real-Time Sentiment Streaming.

Thank You!

Questions?